

CCNA 200-301 v1.1	Security
Banner Commands	
[no]banner login <delimiter><message><delimiter>	Create (or delete) a message users will see as they first log on to the device
[no]banner motd <delimiter><message><delimiter>	Create (or delete) an informational message of the day that users will see after they have logged into the device
banner exec <delimiter><message><delimiter>	Create (or delete) a message users will see when they attempt to enter privileged mode
show running-config include banner	View the current running configuration, showing only banner configurations
Auto Secure Commands	
auto secure	Launch the auto secure wizard; answer the prompts
Password Commands	
enable password [0 7] <password>	Create a Type 7 enable password 0 = unencrypted password will follow 7 = encrypted password will follow
enable secret [0 5] <password>	Create a Type 5 MD5 hashed enable password 0 = unencrypted password will follow 5 = MD5 hash will follow
service password-encryption	Encrypt all current and future clear text passwords with Type 7 encryption
User <name> privilege <level> password secret	Create a user with privilege level and Type 7 or Type 5 password
line console 0	Enter line (console) configuration mode
line vty 0 15	Enter VTY (telnet, SSH) configuration mode
login local	Require the user to log in when connecting
Password Recovery Commands	
confreg 0x2142	Ignore startup-config on bootup
reset	Reboot
confreg 0x2102	Apply startup-config on bootup
show version	Displays hardware and software information about the device including the current configuration register

IPSEC Commands	
crypto isakmp policy <number> crypto isakmp policy 10	Create an IPSEC policy
encryption <algorithm> <strength> encryption aes 256	Declare the encryption algorithm and strength
hash <hashing algorithm> hash sha256	Declare the hashing algorithm
authentication pre-share	Indicate that the IPSEC peers will authenticate via pre-shared key
group <Diffie-Hellman group> group 14	Indicate the key exchange strength (Group 14 - 2048 bit is common)
lifetime <seconds> lifetime 86400	Indicate the lifetime of the security association (SA)
crypto isakmp key <key> address <peer public IP> crypto isakmp key MySecretKey address 33.33.33.33	Set the pre-shared key, indicating the peer public address
crypto ipsec transform-set <name> <encryption> <hash> mode <type>	Configure IKE Phase 2 (IPSEC Transform Set)
crypto ipsec transform-set MY_TRANSFORM_SET esp-aes esp-sha-hmac mode tunnel	
access-list 100 permit ip <subnet for peer 1> <wildcard mask> <subnet for peer 2> <wildcard mask>	Create an ACL specifying interesting traffic (all IP traffic going from subnet 1 to subnet 2)
crypto map <name> <sequence number> ipsec-isakmp crypto map VPN_MAP 10 ipsec-isakmp	Globally enable the VPN map
set peer <peer public IP address> set peer 33.33.33.33	Declare the VPN peer public IP address
set transform-set <name> set transform-set MY_TRANSFORM_SET	Indicate which transform set the map will use
match address <access-list #> match address 100	Indicate which access-list will be used to specify interesting traffic
crypto map <name> crypto map VPN_MAP	Apply the crypto map on the public interface
show crypto isakmp sa	Check phase 1 status
show crypto ipsec sa	Check phase 2 status
ping <IP in peer network>	Test the VPN with a ping
ACL Commands	
access-list [number type <name>] [permit deny] <source> [wildcard mask] access-list 10 permit host 192.168.1.100	Create a standard ACL
access-list [number type <name>] [permit deny] <protocol> <source> [operator <port>] <source-wildcard> <destination> <destination-wildcard> [operator <port>] access-list 100 deny tcp any host 192.168.1.200 eq 80	Create an extended ACL
ip access-group 10 [in out]	Apply ACL on the interface

ACL Verification Commands	
show access-lists	Provide details of all access lists
show access-lists <number name>	Provide details of a specific access list
show access-lists sales	
show access-lists 20	
show ip access-lists	Provide details of IP access lists
show ip access-lists <number name>	Provide details of a specific IP access list
show ip access-lists sales	
show ip access-lists 20	
show ip interface <type><number>	Show IP details of an interface, including ACL and direction
show ip interface g0/0	
show running-config include access-list	Show current running configuration, only include lines starting with "access-list"
DHCP Snooping Commands	
ip dhcp snooping	Enable DHCP snooping globally
ip dhcp snooping vlan <vlan-id>	Enable DHCP snooping on a specific VLAN
ip dhcp snooping vlan 10	
ip dhcp snooping trust	Mark a port as trusted
ip dhcp snooping limit rate <requests per second>	(Optional) Limit DHCP request rate per second
ip dhcp snooping limit rate 5	Requests exceeding this number are dropped On client-facing ports, this is typically set to 5 - 10 On trusted ports, this is typically not set, or set to 1000
show ip dhcp snooping	Display DHCP snooping status
show ip dhcp snooping binding	Display the binding table
DAI Commands	
ip arp inspection vlan <number>	Global: Enable Dynamic ARP Inspection for the VLAN
ip arp inspection vlan 10	
ip arp inspection trust	Configure a trusted interface for DAI
ip arp inspection limit rate <ARP packets per second>	(Optional) Limit ARP messages per second
ip arp inspection limit rate 25	Requests exceeding this number are dropped On endpoint-facing ports, this is typically set from 15 - 25 On trusted ports, this is typically not set, or set to 1000
show ip arp inspection	Check DAI status for VLANs
show ip dhcp snooping binding	Check the DHCP snooping binding table
show ip arp inspection statistics	Check ARP packet statistics
Port Security Commands	
switchport port-security	Enable port security on a specific interface
switchport port-security maximum [number]	Set the maximum number of allowed MAC addresses on a port
switchport port-security maximum 2	
switchport port-security mac-address [mac-address]	Manually specify the MAC address that is allowed in the interface
switchport port-security mac-address 0001.2345.6789	
switchport port-security mac-address sticky	Save the first MAC address heard on the port as the authorized MAC
switchport port-security violation [protect restrict shutdown]	Specify the violation response Default is shutdown
switchport port-security violation protect	

Port Security Verification Commands	
show port-security	Display an overview of port security status for all switchports
show port-security interface <type><number>	Display detailed port security settings for a specific interface
show port-security address	Display secure MAC addresses (learned or static) on the switch
show port-security interface <type><number>	Determine if a specific port has been disabled due to a security violation
show port-security violation	Display a summary of all security violations that have occurred
show mac address-table secure	Display MAC addresses (dynamic and static) that are currently secured on the switch
show running-config interface <type><number>	Display the current configuration of a specific interface, including any security settings that have been applied
Data and Voice Port Security Commands	
switchport voice vlan [vlan ID] switchport voice vlan 7	Configure the switch access port with the voice VLAN ID
switchport port-security maximum [number] switchport port-security maximum 2	Limit the number of allowed MAC addresses on the switchport
switchport port-security violation [protect restrict shutdown] switchport port-security violation protect	Set the violation response on the switchport